$$
\begin{aligned}
&Y_e^{2\,\mathrm{i1\_i2\_}} \to \\
&-y_e^{i1i2} + \frac{1}{16\pi^2}\,\Bigl(\frac{1}{16}\,\left(-6\,\overline{y_e}^{pr}\,y_e^{pi2}\,y_e^{i1r} - y_e^{i1i2}\,\left(g_1^2 + 15\,g_2^2\right)\right) + g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{1,1,0}[m_1, m_{\tilde{e}}^{i2}] - \\
&\quad \frac{1}{2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{e}}^{i2}] + \frac{1}{4}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{1,1,0}[m_1, m_{\tilde{l}}^{i1}] - \\
&\quad \frac{1}{8}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,1,-1}[m_1, m_{\tilde{l}}^{i1}] + \frac{3}{4}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{1,1,0}[m_2, m_{\tilde{l}}^{i1}] - \\
&\quad \frac{3}{8}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{2,1,-1}[m_2, m_{\tilde{l}}^{i1}] + \frac{1}{2}\,\overline{y_e}^{pr}\,y_e^{pi2}\,y_e^{i1r}\,\mathrm{LF}_{1,1,0}[m_{\tilde{e}}^{r}, \tilde{\mu}] - \\
&\quad \frac{1}{2}\,y_e^{i1i2}\,\overline{a_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{1}{2}\,y_e^{i1i2}\,\overline{a_e}^{pr}\,a_e^{pr}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \\
&\quad \frac{1}{2}\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,a_e^{pr} + \overline{a_e}^{pr}\,\left(2\,C_{\phi2^2}\,a_e^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{pr}\right)\right)\,\mathrm{LF}_{3,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \\
&\quad \frac{3}{2}\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,a_e^{pr} + \overline{a_e}^{pr}\,\left(2\,C_{\phi2^2}\,a_e^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{pr}\right)\right)\,\mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \\
&\quad y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_e}^{pr}\,a_e^{pr} + \overline{a_e}^{pr}\,\left(2\,C_{\phi2^2}\,a_e^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{pr}\right)\right)\,\mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \overline{y_e}^{pr}\,y_e^{pi2}\,y_e^{i1r} \\
&\quad\ \mathrm{LF}_{1,1,0}[m_{\tilde{l}}^{p}, \tilde{\mu}] - \frac{3}{2}\,y_e^{i1i2}\,\overline{a_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{3}{2}\,y_e^{i1i2}\,\overline{a_d}^{pr}\,a_d^{pr}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\quad \frac{3}{2}\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,a_d^{pr} + \overline{a_d}^{pr}\,\left(2\,C_{\phi2^2}\,a_d^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_d^{pr}\right)\right)\,\mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \\
&\quad \frac{9}{2}\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,a_d^{pr} + \overline{a_d}^{pr}\,\left(2\,C_{\phi2^2}\,a_d^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_d^{pr}\right)\right)\,\mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\quad 3\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}}\,\tilde{\mu}\,\overline{y_d}^{pr}\,a_d^{pr} + \overline{a_d}^{pr}\,\left(2\,C_{\phi2^2}\,a_d^{pr} + C_{\phi1\phi2}\,\tilde{\mu}\,y_d^{pr}\right)\right)\,\mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\quad \frac{3}{2}\,\tilde{\mu}^2\,y_e^{i1i2}\,\overline{y_u}^{pr}\,y_u^{pr}\,\mathrm{LF}_{2,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{3}{2}\,\tilde{\mu}^2\,y_e^{i1i2}\,\overline{y_u}^{pr}\,y_u^{pr}\,\mathrm{LF}_{3,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \\
&\quad \frac{3}{2}\,\tilde{\mu}\,y_e^{i1i2}\,\left(C_{\phi1\phi2}\,y_u^{pr}\,\overline{a_u}^{pr} + \overline{y_u}^{pr}\,\left(\overline{C_{\phi1\phi2}}\,a_u^{pr} + 2\,C_{\phi2^2}\,\tilde{\mu}\,y_u^{pr}\right)\right)\,\mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\quad \frac{9}{2}\,\tilde{\mu}\,y_e^{i1i2}\,\left(C_{\phi1\phi2}\,y_u^{pr}\,\overline{a_u}^{pr} + \overline{y_u}^{pr}\,\left(\overline{C_{\phi1\phi2}}\,a_u^{pr} + 2\,C_{\phi2^2}\,\tilde{\mu}\,y_u^{pr}\right)\right)\,\mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \\
&\quad 3\,\tilde{\mu}\,y_e^{i1i2}\,\left(C_{\phi1\phi2}\,y_u^{pr}\,\overline{a_u}^{pr} + \overline{y_u}^{pr}\,\left(\overline{C_{\phi1\phi2}}\,a_u^{pr} + 2\,C_{\phi2^2}\,\tilde{\mu}\,y_u^{pr}\right)\right)\,\mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\quad \frac{3}{4}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_1] - \frac{1}{2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{3,1,-2}[\tilde{\mu}, m_1] + 2\,C_{\phi2^2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_1] - \\
&\quad \frac{1}{2}\,m_1\,\tilde{\mu}\,g_1^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] - 4\,C_{\phi2^2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_1] + \\
&\quad \frac{3}{2}\,m_1\,\tilde{\mu}\,g_1^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] + 2\,C_{\phi2^2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_1] - \\
&\quad m_1\,\tilde{\mu}\,g_1^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \frac{9}{4}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_2] - \\
&\quad \frac{3}{2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{3,1,-2}[\tilde{\mu}, m_2] + 6\,C_{\phi2^2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_2] - \\
&\quad \frac{3}{2}\,m_2\,\tilde{\mu}\,g_2^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] - 12\,C_{\phi2^2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_2] + \\
&\quad \frac{9}{2}\,m_2\,\tilde{\mu}\,g_2^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] + 6\,C_{\phi2^2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{5,1,-3}[\tilde{\mu}, m_2] - \\
&\quad 3\,m_2\,\tilde{\mu}\,g_2^2\,y_e^{i1i2}\,\left(\overline{C_{\phi1\phi2}} + C_{\phi1\phi2}\right)\,\mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] - \frac{1}{4}\,\overline{y_e}^{pr}\,y_e^{pi2}\,y_e^{i1r}\,\mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_{\tilde{e}}^{r}] - \\
&\quad \frac{1}{2}\,\overline{y_e}^{pr}\,y_e^{pi2}\,y_e^{i1r}\,\mathrm{LF}_{2,1,-1}[\tilde{\mu}, m_{\tilde{l}}^{p}] - m_1\,g_1^2\,a_e^{i1i2}\,\mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] + \\
&\quad \frac{1}{4}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] - g_1^2\,y_e^{i1i2} \\
&\quad\ \mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] + \frac{1}{4}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] + \\
&\quad \frac{1}{2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{1,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \frac{1}{2}\,C_{\phi2^2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \\
&\quad \frac{1}{2}\,m_1\,C_{\phi1\phi2}\,\tilde{\mu}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] + \frac{1}{4}\,C_{\phi2^2}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
&\quad \frac{1}{4}\,m_1\,C_{\phi1\phi2}\,\tilde{\mu}\,g_1^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \frac{3}{2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{1,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] - \\
&\quad \frac{3}{4}\,C_{\phi2^2}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] + \frac{3}{4}\,m_2\,C_{\phi1\phi2}\,\tilde{\mu}\,g_2^2\,y_e^{i1i2}\,\mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] - \\
&\quad \frac{1}{2}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] + \\
&\quad \frac{1}{2}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] - \\
&\quad \frac{1}{2}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] + \\
&\quad \frac{1}{2}\,m_1\,g_1^2\,\left(C_{\phi2^2}\,a_e^{i1i2} + C_{\phi1\phi2}\,\tilde{\mu}\,y_e^{i1i2}\right)\,\mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}]\Bigr)
\end{aligned}
$$